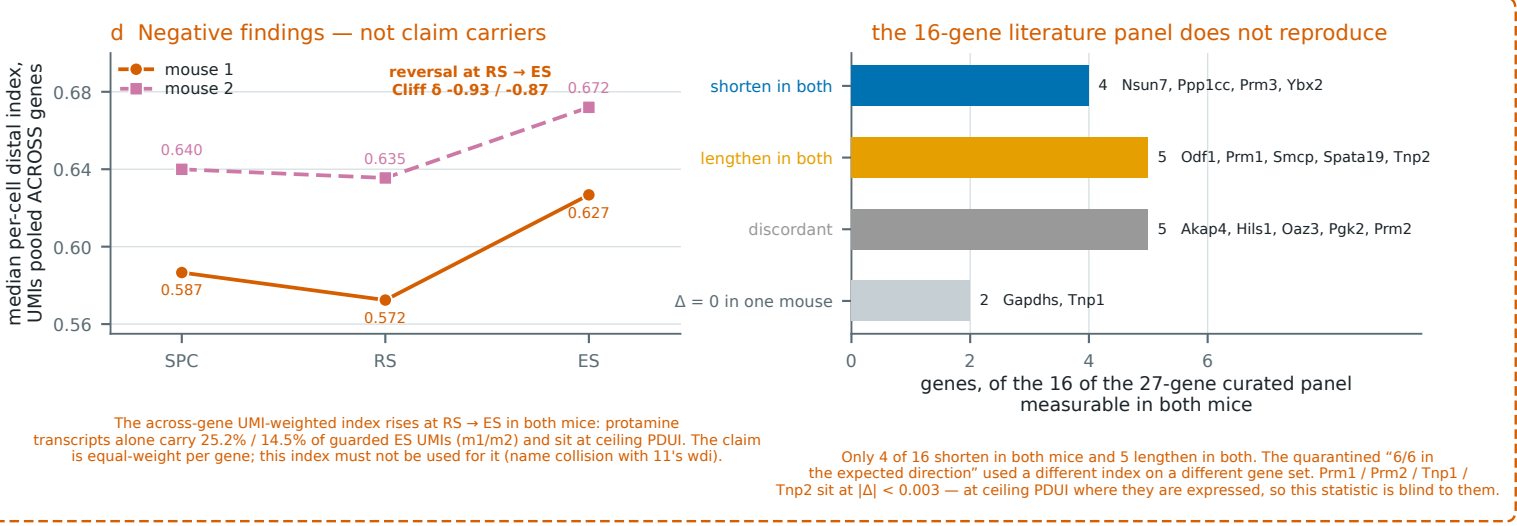
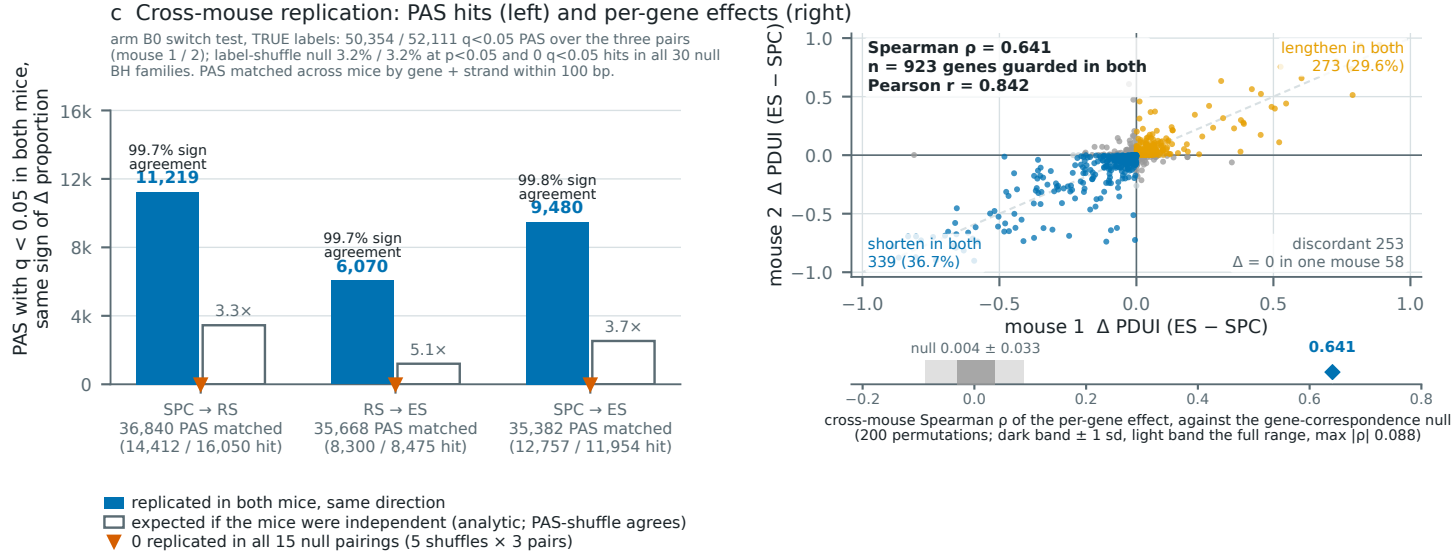
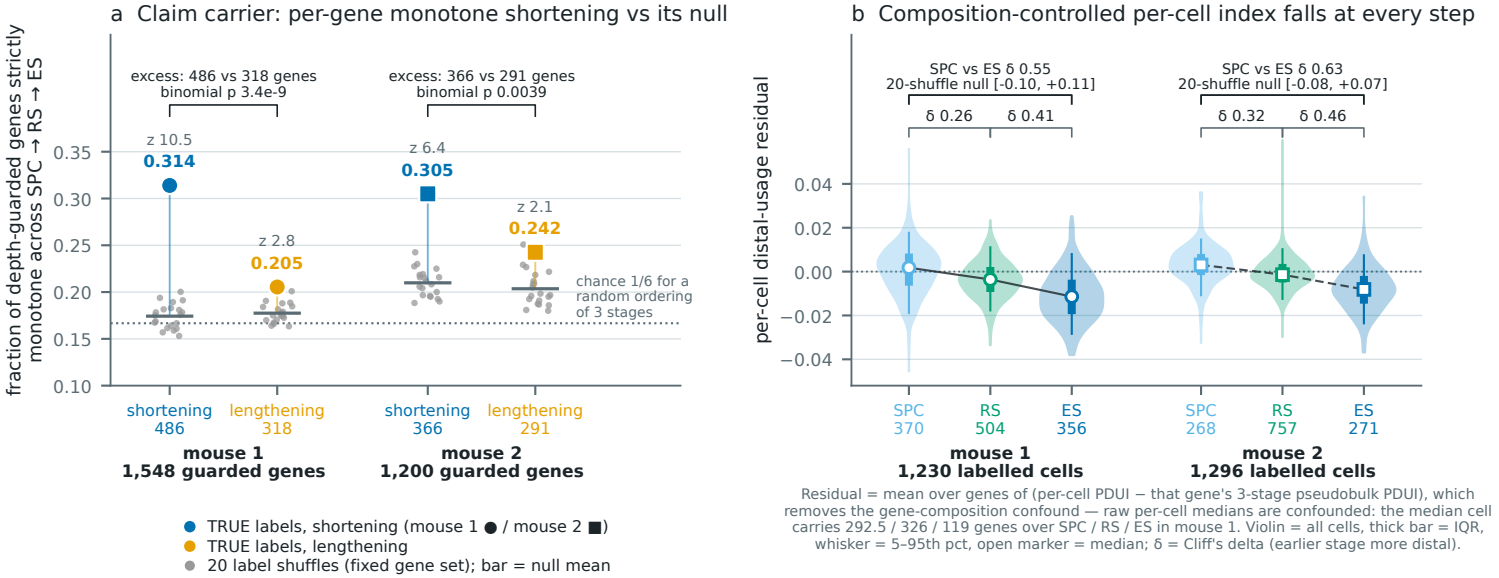


Per-gene 3'UTR usage shifts progressively proximal along spermatocyte → round → elongating spermatid, in both mice

GSE104556 testis, both mice, PeakATail merged caller (frozen tree 9dfdfb3); GEX marker-argmax stage labels; PDUI = distal / (proximal + distal) per gene, pseudobulk per stage, over genes with ≥ 50 UMI at the gene's proximal/distal PAS pair in every stage. 31.4% and 30.5% of guarded genes (486 / 1,548 and 366 / 1,200) shorten monotonically across the three stages, against label-shuffle nulls of 17.4% and 21.0% ($z = 10.5$ and 6.4 , 20 shuffles), exceeding monotone lengthening (486 vs 318 and 366 vs 291 genes); the composition-controlled per-cell distal-usage residual falls at every step (Cliff's δ SPC vs ES = 0.55 and 0.63, outside the whole 20-shuffle null range). This is a per-gene, EQUAL-WEIGHT statement (panel d).



Cautions, binding on any use of this figure. (1) Monotone LENGTHENING is also above its null (z 2.8 / 2.1); the shuffle destroys ordered structure of either sign, so the direction-specific evidence is the EXCESS of shortening over lengthening — strong in mouse 1 (binomial p 3.4e-9), modest in mouse 2 (p 0.0039; 0 of 40 null draws reached the same excess in absolute value). (2) The per-gene MEDIANS are not monotone (RS marginally above SPC in both mice: 0.795 → 0.801 and 0.879 → 0.887); the ordering is carried by the monotone fraction, the paired per-gene tests and the per-cell residual, not by medians, and not by any UMI-weighted index (panel d). Wilcoxon p -values are quoted only next to their shuffle-null column and are therefore not on this figure. (3) SPG is excluded: SPG → SPC LENGTHENS under exactly these marker-argmax labels and rests on 64 / 68 cells (11). (4) Two mice of one study (GSE104556), one protocol — this controls animal-level and sampling noise, not study- or protocol-level artefacts. Stage labels are marker-panel argmax over GEX leiden clusters (98.8% agreement with independent PAS-cluster labels), not ground truth. (5) The ≥ 50 -UMI depth guard keeps 1,548 / 1,200 of $\sim 10,500$ genes with a PAS pair, biased toward highly expressed genes; the null uses the same guarded set. (6) The switch test's null is conservative, not exact (Fisher discreteness), so TRUE hit counts are not "N true switches"; only 5 label shuffles per mouse there (20 for the PDUI arm). Cross-mouse PAS matching is coordinate-based (100 bp, greedy 1:1), so only the matched subset can replicate. (7) RECORD: code is the frozen merged tree 9dfdfb3 (#96 IP-filter/strand + #97), verified SOUND 2026-09-02; the v1 render (4efeb125) stays reproducible via SPERMATOGENESIS_VERSION=v1. Sources: manuscript/18 v2 section (verdict SOUND, 2026-09-02); v1 retained as the labelled comparison and results/stage3_spermatogenesis_v2/ (REPORT_PROVISIONAL.md §2-4, summary/ and mouse{1,2}/summary/). Every plotted value: results/figures/manuscript/fig5_spermatogenesis*.tsv.